Comprehensive Efforts in Computer Vision Projects

Progress, Achievements, and Challenges

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Date: 22-10-24

Agenda

- Overview of Key Projects
- Progress and Achievements
- Challenges and Solutions
- Improvements and Learning
- Next Steps and Future Plans

Tomato Maturity Classification

Data Split:

- Training: 576 images
- Validation: 144 images
- Test: 180 images
- Unseen: 200 images

Models Used: DenseNet121 and MobileNetV2

- DenseNet 121 performed better; MobileNetV2 showed overfitting

Results:

- 100% training accuracy, 99% validation accuracy
- 98% accuracy on both test and unseen datasets

Tomato Instance Segmentation

Model: YOLOv8 and YOLO11n for tomato instance segmentation

Dataset: 400 images for training, 100 images for validation (Roboflow)

Use Cases: Tomato quality grading automation with the segmented images.

Observations: Performance and potential improvement areas for raw tomatoes.

Classification Model: Trained a tomato maturity classification model with segmented images which gave 96% accuracy.

Object Detection Experiment

Objects Detected: Hair, feathers, plastic items, and worms. Trained a common YOLO segmentation model for these objects

Datasets Used: Various datasets from Roboflow

Potential Applications: Quality control in food processing

Cuts and Damages Model

Motivation: Found that detached head or completely damaged head is not getting detected in the cuts and damages model.

Model: Trained a cuts and damages model with different number of sardine images in the range of 100 to 300 with different number of epochs and different versions of YOLO v8 model. The accuracy was low due to annotating even the minute damages in the dataset.

Research Paper Approach Experimentation

Techniques Used: Feature extraction with VGG19 combined with ML models (Logistic Regression, SVM, etc.)

Experiments Conducted:

- Different datasets and augmentations (FFE data, warehouse data)
- 2-class (Good/Bad) and 3-class models

Outcomes: Poor results compared to existing models.

Sardine Eye Freshness Classification

- **Dataset Development:** Created sardine fish eye dataset.
- **Eye Detection:** Trained a YOLO segmentation model for fish eyes. Cross-tested with other species (mackerel, tilapia, Indian salmon). Worked well only for Sardine and Mackerel.
- **Experiments:** Trained CNN model for freshness classification. The accuracy was 69%.

Data Management and Workflow Improvements

Google Drive Cleanup:

- Removed unused files to optimize storage

Data Collection Enhancements:

- Created automated data collection and visualization tools
- Implemented consistent updates to data files

Thank You!